

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	7 March 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	EV Data Collection	Collect EV charging data such as time, location, duration and power consumed Import battery and vehicle performance datasets
FR-2	Charging Pattern Analysis	Analyze charging behavior by time of day identify peak charging hours and high demand area
FR-3	Battery Performance Monitoring	Track battery metrics such as stste of charge(SOC) analyze depth of discharge and charging cycles
FR-4	Driving Range Analysis	Compare predicted vs actual driving range identify vehicles low efficiency
FR-5	EV Model Comparison	Compare efficiency of different EV models analyze cost per kilometer and charging cost
FR-6	Data Visualization Dashboard	Create interactive dashboards using Tableau Heatmaps, scatter plots, bar charts for charging patterns, battery health, and model comparisons

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple and interactive so users can easily analyze EV data.
NFR-2	Security	Ensure secure access to EV data and protect user information.

NFR-3	Reliability	The system should consistently provide accurate visualizations and analysis.
NFR-4	Performance	The system should process EV datasets and generate dashboards quickly.
NFR-5	Availability	The visualization tool should be accessible whenever users need to analyze EV data.
NFR-6	Scalability	The system should support increasing EV datasets and multiple users without performance issues.